1 Appendix for Section 4

1.1 Apparent Balanced Growth in $\bar{c}$ and $\mathrm{cov}(c, \boldsymbol{p})$

{sec:ApndxBalance

Section 4.2 demonstrates some propositions under the assumption that, when an economy satisfies the GIC, there will be constant growth factors $\Omega_{\bar{c}}$ and Ω_{cov} respectively for $\bar{c}$ (the average value of the consumption ratio) and $\mathrm{cov}(c, \boldsymbol{p})$. In the case of a Szeidl-invariant economy, the main text shows that these are $\Omega_{\bar{c}} = 1$ and $\Omega_{\mathrm{cov}} = \mathcal{G}$. If the economy is Harmenberg- but not Szeidl-invariant, no proof is offered that these growth factors will be constant.

1.2 $\log c$ and $\log(\mathrm{cov}(c, \boldsymbol{p}))$ Grow Linearly

Figures 1 and 2 plot the results of simulations of an economy that satisfies Harmenberg- but not Szeidl-invariance with a population of 4 million agents over the last 1000 periods (of a 2000 period simulation).[1] The first figure shows that $\log \bar{c}$ increases apparently linearly. The second figure shows that $\log(-\mathrm{cov}(c, \boldsymbol{p}))$ also increases apparently linearly. (These results are produced by the notebook `ApndxBalancedGrowthcNrmAndCov.ipynb`).

[1] For an exposition of our implementation of Harmenberg's method, see this supplemental appendix.

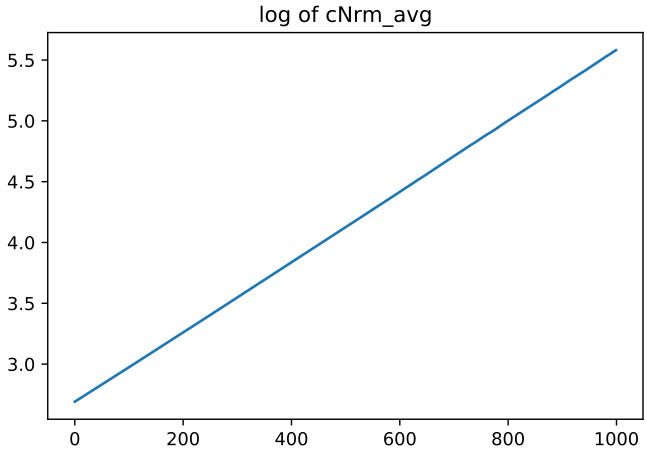

Figure 1 Appendix: $\log\ \mathfrak{c}$ Appears to Grow Linearly {fig:logcNrm}

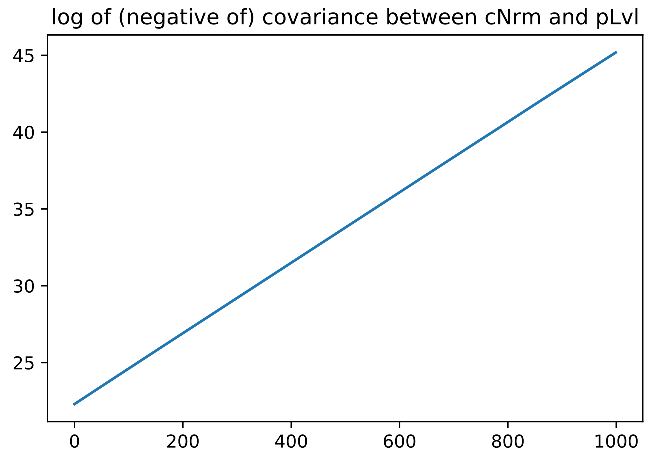

Figure 2 Appendix: $\log\left(-\operatorname{cov}(c, \boldsymbol{p})\right)$ Appears to Grow Linearly {fig:logcov}

(See (2))

Table 1 Model Characteristics Calculated from Parameters

{table:Calibration}

Description	Symbol and Formula			Approximate Calculated Value
Finite Human Wealth Factor	$\tilde{\mathcal{R}}^{-1}$	$\equiv$	$\mathcal{G}/\mathsf{R}$	0.990
PF Value of Autarky Factor	$\beth$	$\equiv$	$\beta\mathcal{G}^{1-\gamma}$	0.932
Growth Compensated Permanent Shock	$\underline{\psi}$	$\equiv$	$(\mathbb{E}[\psi^{-1}])^{-1}$	0.990
Uncertainty-Adjusted Growth	$\underline{\mathcal{G}}$	$\equiv$	$\mathcal{G}\underline{\psi}$	1.020
Utility Compensated Permanent Shock	$\underline{\underline{\psi}}$	$\equiv$	$(\mathbb{E}[\psi^{1-\gamma}])^{1/(1-\gamma)}$	0.990
Utility Compensated Growth	$\underline{\underline{\mathcal{G}}}$	$\equiv$	$\mathcal{G}\underline{\underline{\psi}}$	1.020
Absolute Patience Factor	Þ	$\equiv$	$(\mathsf{R}\beta)^{1/\gamma}$	0.999
Return Patience Factor	$\text{Þ}/\mathsf{R}$	$\equiv$	$\text{Þ}/\mathsf{R}$	0.961
Growth Patience Factor	$\text{Þ}/\mathcal{G}$	$\equiv$	$\text{Þ}/\mathcal{G}$	0.970
Modified Growth Patience Factor	$\text{Þ}/\mathcal{G}\mathbb{E}[\psi^{-1}]$	$\equiv$	$\text{Þ}/\underline{\mathcal{G}}$	0.980
Value of Autarky Factor	$\underline{\underline{\beth}}$	$\equiv$	$\beta\mathcal{G}^{1-\gamma}\underline{\underline{\psi}}^{1-\gamma}$	0.941
Weak Return Impatience Factor	$\wp^{1/\gamma}\text{Þ}$	$\equiv$	$(\wp\beta\mathsf{R})^{1/\gamma}$	0.071